ELSEVIER

Soil Dynamics and Earthquake Engineering 19 (2000) 533–547

SOIL DYNAMICS AND EARTHQUAKE ENGINEERING

www.elsevier.com/locate/soildyn

Local transmitting boundaries for transient elastic analysis

L. Kellezi*

Department of Structural Engineering and Materials, Denmark Technical University, DK2800 Lyngby, Denmark

Received 1 March 1999; accepted 9 February 2000

Abstract

The aim of this paper is to investigate and develop alternative methods of analyzing problems in dynamic soil–structure-interaction (SSI). The interaction means that the amplitude of structural response is effected by additional energy dissipation through radiation and material damping in the soil. The surrounding soft soil behaves as a natural damper for a massive and stiff structure supported or embedded in it. The main focus is the major difficulty posed by such an analysis — the phenomena of waves that radiate outward from the excited structures towards infinity. In numerical calculations only a finite region of the foundation medium is analyzed and something is done to prevent the outgoing radiation waves from reflecting at the boundary region.

Development of a simple and efficient finite element (FE) procedure for the solution directly in the time domain of transient SSI problems is the main concern. The central feature of the procedure is local absorbing boundaries used to render the computational domain finite. These boundaries are local in both time and space and are completely defined by a pair of symmetric stiffness and damping matrices. As the effort for implementing them is the same as for the impedance boundary condition (BC) considering the angle of incidence, standard assembly procedure can be used. Due to the local nature they also preserve the overall structure of the global equations of motion. Even though the focus is in the time domain the same equations of motions can be used to determine the solution under time-harmonic excitation directly in the frequency domain. Explicit formulae for the element matrices are included in the paper and numerical examples for transient radiation model problems to illustrate the validity and accuracy of the new procedures, are given. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Dynamic soil–structure-interaction; Transmitting boundary condition; Finite element method; Radiation condition; Cone models; Linear cone models; Axisymmetric analysis; Plain strain analysis; Rayleigh damping

1. Introduction

The well-recognized numerical problem in dynamic soil–structure-interaction (SSI) analysis is how to simulate computationally the far-field medium at infinity. Based on the radiation condition, at a sufficient distance from the source only outgoing waves are present, no incoming waves propagating from infinity towards the structure exist when such sources of energy are avoided. This radiation condition leads to a boundary-value problem for an unbounded domain with a unique solution. To analyze the semi-infinite domain of the soil numerically, a surface or zone is chosen called *interaction horizon* [1]. The constitutive associated with the nodes on this horizon represents the significant features of the far field. The interaction horizon can be situated depending on the problem. Two extremes are possible [2]: the interaction horizon can be the generalized structure–soil interface (substructure method), Fig.1a, or the interaction horizon can be identical to an artificial boundary up to which the soil is modeled with e.g. FEs (direct method), Fig. 1b.

The substructure method, based on the fundamental solutions, which satisfy exactly the radiation condition formulated at infinity and on the principle of superposition, applies only to regular linear halfspace models. On the other hand, the direct method models numerically the structure and a region of soil in contact up to the artificial boundary. The rigorous BC is global in space and time and is described through integro-differential operators. As this is computationally expensive and in the direct method artificial boundary is placed far from the energy source, approximate or local BCs can be formulated using only differential operators with respect to space and time. Besides modeling the soil's stiffness up to infinity, reflections of the outgoing propagating waves on the artificial boundary must be avoided also. Therefore, the artificial boundary must act as a *transmitting, absorbing or nonreflecting boundary*. If the transmitting boundary surface is denoted with Γ_∞ where subscript ∞ is used to indicate that this artificial boundary

* Present address: Building Research Institute, Department of Production Engineering, Tatehara 1, Tsukuba, Ibaraki 305-08-02, Japan. Tel.: +81-298-760-697; fax: +81-298-646-774.

E-mail address: lkellezi@yahoo.com (L. Kellezi).

0267-7261/00/$ - see front matter © 2000 Elsevier Science Ltd. All rights reserved.
PII: S0267-7261(00)00029-4

Nomenclature

Greek letters

α	The predicted angle of wave incidence (rad)
α_0	Rayleigh damping coefficient (–)
β_0	Rayleigh damping coefficient (–)
Γ	Notation to indicate the boundary surface (–)
γ	Shear strain component (N/m^2)
Δt	Time step (s)
Δx	Spatial step in x-direction (m)
η	Normal coordinate for z-direction (m)
θ	Wilson coefficient (–)
κ	Correction coefficient (–)
μ	Shear modulus (N/m^2)
ν	Poisson's ratio (–)
ξ_s	Soil material damping ratio (–)
ρ	Density (kg/m^3)
σ	Stress component (N/m^2)
$\{\sigma\}$	Stress vector from the transmitting boundary (N/m^2)
τ	Tangential stress component (N/m^2)
ϵ	Normal strain component (–)
χ_R	Wavelength of R-waves (m)
χ_s	Wavelength of S-waves (m)
Ω	Notation to indicate the medium (–)
ω	Circular frequency of excitation (rad/s)
∞	Infinity extend (–)
∂	Partial derivative (–)

Roman letters

c_p	Wave velocity for P-wave (m/s)
c_R	Wave velocity for R-wave (m/s)
c_s	Wave velocity for S-wave (m/s)
$[\boldsymbol{C}]$	Damping matrix (Ns/m)
$[\boldsymbol{D}_C]$	Constitutive damping matrix (Ns/m)
$[\boldsymbol{D}_K]$	Constitutive stiffness matrix (Ns/m^2)
E	Young's Modulus (N/m^2)
f	Function identification (–)
$[\boldsymbol{M}]$	Mass matrix (kg)
N_i	FE shape functions (–)
$[\boldsymbol{N}]$	Transformation matrix (–)
$\mathbf{n}$	Vector normal to the boundary (–)
r	Radial coordinate (m)
$\mathbf{r}$	Wave direction vector (–)
s	Velocity ratio equal to $\sqrt{2(1-\nu)/(1-2\nu)}$ (–)
t	Time
T_p	Predominant period of excitation (s)
u	Displacement in x-direction (m)
$u_{,t}$	Velocity in x-direction (m/s)
v	Displacement in y-direction (m)
w	Displacement in z-direction (m)
W	Displacement amplitude in z-direction (m)
x	Horizontal Cartesian coordinate (m)
x_b	Horizontal coordinate of the boundary (m)
y	Out-of-plane horizontal Cartesian coordinate (m)
z	Vertical Cartesian coordinate (m)
z_b	Vertical coordinate of the boundary (m)

will simulate infinity, for any point with coordinates x in this surface the BC in terms of stresses can be given as

$$\sigma(x,t) = -\mathscr{F}_\infty[u(x,t), u_{,t}(x,t)] \tag{1}$$

where σ represents stresses at the boundary location x, t time, u displacement and $u_{,t}$ velocity at the same location. $\mathscr{F}_\infty$ is a function representing material properties of the infinite domain, location and the geometry of the boundary surface Γ_∞. When surface Γ_∞ is located and $\mathscr{F}_\infty$ is determined the total system becomes finite and can be solved by ordinary numerical techniques. In determining function $\mathscr{F}_\infty$, usually it is assumed that the infinite domain behaves linearly.

Numerical procedures for the dynamic SSI analysis may be classified according to the nature of the time dependence as either time-harmonic or transient. In the linear case, time-harmonic solutions can be used indirectly employing Fourier transform to solve the transient problems. Still a direct time integration approach is necessary whenever nonlinearities occur and may be advantageous for some classes of linear problems. For example, in linear problems exhibiting broadband phenomena, the indirect approach may not be computationally feasible. Also, the measurements of actual performance of SSI problems are usually recorded directly in the time domain, so it may be of interest to use or compare this information with that predicted directly by the mathematical models. In this paper local transmitting boundaries applicable for transient analysis are considered.

2. Local transmitting boundary techniques

A short description of most popular local transmitting boundaries for time domain analysis which can be considered as different approaches of the function $\mathscr{F}_\infty$ in Eq. (1) is given here. Some of these schemes can have both applications for harmonic or transient analysis. This depends on the method of implementation of the boundary. A detailed analysis of these techniques including different areas of application is given by Givoli [3] who in other words in frequency domain called the function $\mathscr{F}_\infty$ a DtN-map: a Dirichlet to Neumann map describing the stress–displacement relationship at the boundary for any frequency where a 'Dirichlet item', the displacement, is mapped to a 'von Neumann item', the stress.

The simplest local absorbing BC is the classical normal impedance or *standard viscous boundary* [4]. Its performance is known to deteriorate as the position approaches the source of perturbation especially in the low-frequency limit. The great advantage of this solution is that absorption characteristics are independent of frequency and it can easily be implemented in FE codes operating in time as well as in the frequency domain, although it fails for static loads since dashpots have no static stiffness. Another approach in this context was presented by White et al. [5] on the basis of a systematic formulation applicable also to

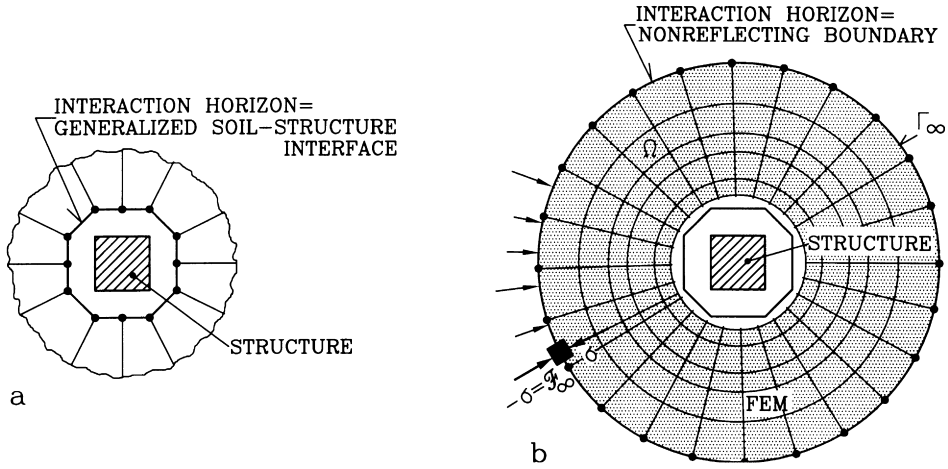

Fig. 1. Dynamic models. (a) Substructure method. (b) Direct method.

anisotropic materials. Here the dashpot coefficients are given as functions of Poisson's ratio ν and the solution is known as *unified viscous boundary*. Later Akiyoshi [6] proposed a viscous boundary for shear waves known as *compatible viscous boundary*. This solution has the disadvantage of involving a convolution integral in its formulation losing the local character. Smith [7] proposed *superposition boundary* based on the principle of the virtual image. The analytical formulation is exact and independent of frequency and incidence angle; however, superposition of solutions requires the system to behave linearly and 2^m solutions are necessary for m number of dimensions. This scheme was improved by Kunar and Marti [8] overcoming the problem of multiple reflections.

A general transmitting boundary was proposed by Lindman [9] who used projection operators to obtain BCs of high degree of effectiveness for the scalar wave equation. Closely related to this proposal a systematic method for obtaining a hierarchy of local BCs was presented by Engquist and Majda [10] and Clayton and Engquist [11,12] who derived a sequence of BCs of increasing order using rational approximations to pseudo-differential operators. They were based on the original idea of constructing equations which transmit waves in one direction. It was shown that these operators called *paraxial boundary* provide well-posed conditions. The high-order approximations involve progressively more points near the boundary losing their local character. Reynolds [13] derived the paraxial boundary equations making another factorization of the scalar wave differential operator. These boundary schemes, which are a generalization of the viscous boundary, are usually applied by finite differences and cannot readily be implemented in the FE calculations. Cohen and Jennings [14] extended the paraxial boundary applicable for FE calculations. The FE treatment of the first-order terms leads to nonsymmetric matrices causing spurious oscillations. To remove these oscillations the 'upwinding' of nonsymmetric paraxial terms is necessary and an interfacing effect caused by superposition of the boundary integrals of the elastic and paraxial equations should be considered, which makes the boundary cumbersome, and the response is not better than viscous boundary. This was called an *extended paraxial boundary*. Bayliss and Turkel [15] have introduced another sequence of boundary operators based on the asymptotic expansion of the solution for the acoustic radiation problem in negative powers of the radial distance from the source and the acoustic wave number. When applied to an artificial boundary the operators in this sequence provide an increase in accuracy but also in difficulty of implementation. This is known as *Bayliss and Turkel boundary*. Liao and Wong [16] formulated an explicit-time-integration scheme in connection with *extrapolation algorithm* based on polynomial fitting. It is reported also by Cheng and Cheng [17] that this boundary according to the number of terms is another way of formulating the hierarchy of paraxial boundaries. This scheme is only applicable for explicit time domain analysis. Underwood and Geers [18] derived the doubly asymtotic (DA) *approximation* which is asymptotically exact at high- and low-frequency limits and the resulting scheme is local in time though not in space. In the high-frequency limit the infinite medium appears to the boundary as an array of dashpots and at low frequency as an array of springs. As the boundary element method is used to define the static stiffness coefficients, they are fully coupled and nonsymmetric. So this boundary is temporarily local but spatially global. The advantage is that no permanent displacements exist anymore in the elastic system as in the case of viscous boundary. Keys [19] derived BCs for the acoustic wave equation based on a different decomposition of the wave equation consisting of a vectorial representation of incoming and outgoing waves. He did not approximate the outgoing component but the direction in which the waves propagate was considered. Using another method, Higdon [20–22] derived more complicated BCs constructed by concatenating several of these operators for acoustic and

elastic media. This is known as *multi-directional (MD) boundary*. When the number of operators goes to infinity this gives a global formulation; however, in practice, only the product of two or three operators is taken. In general, the optimal angle of incidence depends on the considered problem. If the wave motion is created from a localized source, then this angle at each boundary point is well defined. As elastic waves are more complicated than acoustic waves, this boundary is applied to each component of the displacement vector and it can be tuned to absorb different types of waves. The disadvantages of this boundary are that it does not cover the static case, evanescent waves, and it is not suitable for FE implementations. Finally, Wolf and Song [23] formulated a *doubly asymptotic multi-directional (DAMD)* boundary which combines the advantages of DA approximation and MD formulation. Here, the MD outward plane wave BC, which consists of the product of two operators, is formulated for the interaction forces and not for the displacements and is discretized using forward difference formulas. The static or low-frequency limit is modeled employing a DA boundary which is implemented implicitly. Stability is guaranteed for Courant numbers less than one. The disadvantage of this boundary is that because of DA approximation the static stiffness matrix is fully coupled and nonsymmetric. To make the boundary local in space an approximated banded static stiffness matrix is used. It is reported that *the accuracy is governed by the static behavior and to increase it the boundary should be situated quite far from the energy source*.

At first glance all transmitting boundaries applicable for time domain analysis appear to be different in mathematical formulation, physical perception and numerical implementation. However, it is reported in Refs. [23,24] and others that they are all mathematically related.

3. The proposed transmitting boundaries

From the above investigation of local boundary schemes it is noted that boundary operators of order two or more give nonsymmetric matrices when they are implemented in the FE method. On the other hand, the accuracy of the boundary is governed by the static behavior [23]. So, the question arises as how to formulate local transmitting boundaries accurate also for low frequency using first-order operators. In this section, based on the preselected wave fields, local boundaries of the first order for two-dimensional (2D) and three-dimensional (3D) dynamic analysis are formulated.

When an impulse is acting on an elastic halfspace medium the energy is radiated by shear SV- and SH-waves with velocity c_s, dilatational P-waves with velocity c_p and surface Rayleigh R-waves with velocity c_R. In order for the waves to transmit energy at infinity, the displacement amplitude must die off at large distances in a special low. A radiation criterion states that radiation of energy occurs when the displacement amplitude decays at infinity in inverse proportion to the square root of the surface area at infinity. For 2D plane strain analysis, the surface at infinity for body waves can be considered as an infinitely long cylinder with radius $r \rightarrow \infty$. For R-waves the surface at infinity is an infinitely long vertical plane with height equal to one wavelength of the surface waves. In 3D analysis, the surface at infinity for body waves is a large hemisphere with radius $r \rightarrow \infty$. For R-waves the surface at infinity is a flat cylinder with radius r and height approximately one R-wave length χ_R.

From the radiation criterion, in the plane strain case body waves can be considered to decrease in proportion to $1/\sqrt{r}$ and surface waves should propagate with constant amplitude. In the 3D case the amplitude of the body waves can be approximated to decrease in proportion to the ratio $1/r$ where r is the distance from the input source and the amplitude of the R-waves in proportion to $1/\sqrt{r}$ where r is the radius of the cylindrical surface. These ideas are developed below.

3.1. 2D plane strain analysis

In the context of the one-dimensional (1D) wave theory let us consider a longitudinal and a shear transient wave pattern of arbitrary shapes with constant amplitudes propagating in the positive x-direction:

$$u(x,t) = f(x - c_p t) \tag{2}$$

$$w(x,t) = f(x - c_s t) \tag{3}$$

The viscous boundary was derived in [4] by considering the propagation of harmonic waves, but another method could be by considering transient waves (2) or (3) and deriving for each of them stress–strain relationships in the context of the 1D wave theory. These will derive the well-known boundary differential equations:

$$\left(\frac{\partial}{\partial t} + c_p \frac{\partial}{\partial x}\right) u = 0 \tag{4}$$

$$\left(\frac{\partial}{\partial t} + c_s \frac{\partial}{\partial x}\right) w = 0 \tag{5}$$

These equations represent the normal impedance or standard viscous boundary for waves propagating in the positive x-direction, often applied to the artificial boundary for 2D and 3D dynamic analysis. By multiplying Eqs. (4) and (5) with the complementary operators, respectively, equations of motion for the rod models result.

If the angle of incidence is considered, which is the angle between the wave direction represented by the unit vector $\boldsymbol{r}$ and outward unit vector $\boldsymbol{n}$, normal to the boundary surface, then Eqs. (4) and (5) take the form derived in [20]:

$$\left(\cos\alpha \frac{\partial}{\partial t} + c_p \frac{\partial}{\partial x}\right) u = 0 \tag{6}$$

$$\left(\cos\alpha\frac{\partial}{\partial t}+c_s\frac{\partial}{\partial x}\right)w=0 \tag{7}$$

where α is the angle between $\boldsymbol{n}$ or the positive x-direction and $\boldsymbol{r}$. If the coordinates of these unit vectors are denoted as $\boldsymbol{n}(n_x, n_z)$ and $\boldsymbol{r}(r_x, r_z)$, respectively, for an (x, z) coordinate system in plane strain analysis, the stress components at the boundary considering the angle of incidence can be given in general as:

$$\{\sigma\}=[\boldsymbol{D}_C]\{u_{,t}\} \tag{8}$$

where a comma notation is used for the time derivative. The constitutive damping matrix $[\boldsymbol{D}_C]$ can be given as:

$$[\boldsymbol{D}_C]=\rho c_p(\boldsymbol{n}\cdot\boldsymbol{r})[\boldsymbol{N}]+\rho c_s(\boldsymbol{n}\cdot\boldsymbol{r})\{[\boldsymbol{I}]-[\boldsymbol{N}]\} \tag{9}$$

where ρ is density and the transformation matrix $[\boldsymbol{N}]$ is equal to

$$[\boldsymbol{N}]=\begin{bmatrix} n_x^2 & n_xn_z \\ n_zn_x & n_z^2 \end{bmatrix} \tag{10}$$

$[\boldsymbol{I}]$ is the identity matrix. So Eq. (8) is the viscous boundary considering the angle of incidence.

As mentioned before, body waves in plane strain analysis should propagate radially outward along a cylindrical wavefront. In the context of the 1D wave theory, a cylindrical P- or S-wave travelling in the positive z-direction can be closely approximated by:

$$u(z,t)=\frac{1}{\sqrt{z}}f(z-c_st) \tag{11}$$

$$w(z,t)=\frac{1}{\sqrt{z}}f(z-c_pt) \tag{12}$$

In a similar way as for the rod models, the boundary differential equations in these cases can be obtained from the stress and strain relationships. The constitutive relation at any point z, for example for travelling S-wave, is expressed as:

$$\tau(z,t)=\mu\gamma=\rho c_s^2u_{,z}=\rho c_s^2\left(-\frac{1}{2\sqrt{z^3}}f+\frac{1}{\sqrt{z}}f'\right) \tag{13}$$

where τ is shear stress, μ shear Modulus and γ shear strain.

The particle velocity is equal to:

$$u_{,t}(z,t)=-\frac{c_s}{\sqrt{z}}f'(z-c_st) \tag{14}$$

Comparing Eqs. (11), (13) and (14) the boundary stress is given as

$$\tau(z,t)=-\left[\frac{\rho c_s^2}{2z}u(z,t)+\rho c_su_{,t}(z,t)\right] \tag{15}$$

As differential equations satisfied respectively from the outgoing S- or P-pulses the above equation can be written as

$$\left[\frac{\partial}{\partial t}+\frac{c_s}{2z}+c_s\frac{\partial}{\partial z}\right]u=0 \tag{16}$$

$$\left[\frac{\partial}{\partial t}+\frac{c_p}{2z}+c_p\frac{\partial}{\partial z}\right]w=0 \tag{17}$$

To use Eqs. (16) and (17) as absorbing boundary for body waves, the shape of the 1D model should be defined. As the rod wave equation can be written as product of two complementary boundary operators the same thing is done to derive the differential equation of the model. So the product in the case of shear waves could be:

$$\left[\frac{1}{c_s}\frac{\partial}{\partial t}+\left(\frac{1}{2z}+\frac{\partial}{\partial z}\right)\right]\left[\frac{1}{c_s}-\left(\frac{1}{2z}+\frac{\partial}{\partial z}\right)\right]u=0 \tag{18}$$

Disregarding the term $1/4z^2$, which grows smaller as z increases, the differential equation of a horn with linear area variation results as below [25]:

$$\frac{1}{c_s}u_{,tt}-\frac{1}{z}u_{,z}-u_{,zz}=0 \tag{19}$$

So the missing part of the linear cones from the boundary location z to infinity is modeled by a mechanical system which consists of a spring and a damper with frequency-independent coefficients, Eq. (15). The stiffness terms are in inverse proportion to the double-apex axis of the models. These models can be used as transmitting boundary for body waves in plane strain analysis arranged as in Fig. 2, on the left. Only a cross-section of the 2D and 3D problems are given. The stress BC equals:

$$\{\sigma\}=[\boldsymbol{D}_K]\{u\}+[\boldsymbol{D}_C]\{u_{,t}\} \tag{20}$$

where constitutive stiffness matrix $[\boldsymbol{D}_K]$ is given as

$$[\boldsymbol{D}_K]=\frac{\rho c_p^2}{2r}(\boldsymbol{n}\cdot\boldsymbol{r})[\boldsymbol{N}]+\frac{\rho c_s^2}{2r}(\boldsymbol{n}\cdot\boldsymbol{r})\{[\boldsymbol{I}]-[\boldsymbol{N}]\} \tag{21}$$

and constitutive damping matrix $[\boldsymbol{D}_C]$ is given by Eq. (9). In Eq. (21) r is the distance from the boundary node to the source location which, for a circular boundary surface, is constant for all the boundary nodes and the scalar product of $\boldsymbol{n}$ and $\boldsymbol{r}$ is equal to one. For any other boundary shape r should be given for each node or each Gauss point together with the unit vectors $\boldsymbol{n}$ and $\boldsymbol{r}$. So, for plane strain analysis Eq. (20) can be employed as the transmitting boundary in the area where body waves propagate. On the other hand, Eq. (9) in the form:

$$[\boldsymbol{D}_C]=\rho sc_R(\boldsymbol{n}\cdot\boldsymbol{r})[\boldsymbol{N}]+\rho c_R(\boldsymbol{n}\cdot\boldsymbol{r})\{[\boldsymbol{I}]-[\boldsymbol{N}]\} \tag{22}$$

where s is the ratio of P- to S-wave velocities equal to $s=\sqrt{2(1-\nu)/(1-2\nu)}$, can be employed as lateral boundary at a depth equal to one χ_R based on the radiation criterion. The wavelength χ_R is constant for a harmonic pulse and varies for a transient one. In the transient case, the predominant frequency of the pulse should be considered to determine the length of the lateral boundary where surface waves will be absorbed.

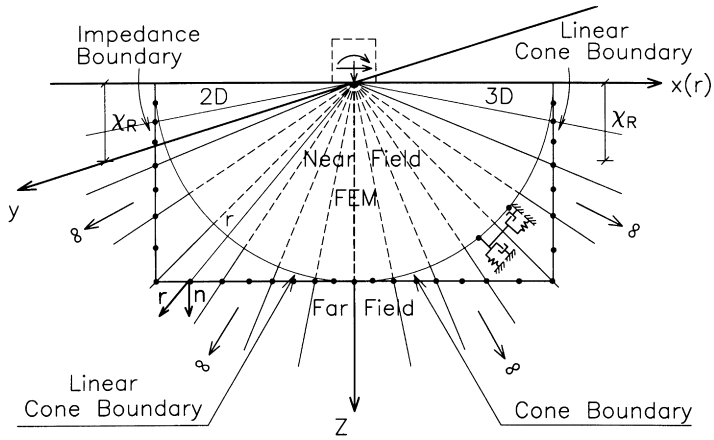

Fig. 2. Local transmitting boundaries for 2D and 3D analysis.

A transmitting boundary for R-wave absorption was developed in [4] based also in the 1D wave propagation using a travelling wave of constant amplitude and was corrected by stress conditions of elasticity equations for homogeneous halfspace, in-plane motion. However this boundary algorithm is frequency-dependent.

3.2. 3D axisymmetric analysis

As body waves in the 3D case propagate with a hemispherical wavefront, a wave pattern with amplitude in proportion to $1/r$ could be considered based on the radiation criterion. In the context of 1D models the outgoing wave patterns for example, for a P-wave, can be given as

$$w(z,t) = \frac{1}{z} f(z - c_p t) \tag{23}$$

From the stress–strain relation, at a cross-section of the model with coordinate z:

$$\sigma(z,t) = E\epsilon(z,t) = \rho c_p^2 w_{,z} = \rho c_p^2 (-\frac{1}{z^2} f + \frac{1}{z} f') \tag{24}$$

where σ is the normal stress, E Young's Modulus and ϵ normal strain. The velocity is then given as:

$$w_{,t}(z,t) = -\frac{c_p}{z} f'(z - c_p t) \tag{25}$$

Comparing Eqs. (23)–(25), the stress at the boundary location is given as a function of the displacement and velocity as:

$$\sigma(z,t) = -\left[\frac{\rho c_p^2}{z} w(z,t) + \rho c_p w_{,t}(z,t) \right] \tag{26}$$

As a differential equation the absorbing boundary for this 1D model can be written as:

$$\left[\frac{\partial}{\partial t} + \frac{c_p}{z} + c_p \frac{\partial}{\partial z} \right] w = 0 \tag{27}$$

The equation of motion for the model can be determined in the same way as for the linear cone model by writing the product:

$$\left[\frac{1}{c_p}\frac{\partial}{\partial t} + \left(\frac{1}{z} + \frac{\partial}{\partial z}\right) \right] \left[\frac{1}{c_p}\frac{\partial}{\partial t} - \left(\frac{1}{z} + \frac{\partial}{\partial z}\right) \right] w = 0 \tag{28}$$

which, neglecting the residual term $1/z^2$, gives the differential equation of a horn with conical area variation [25]

$$\frac{1}{c_p} w_{,tt} - \frac{2}{z} w_{,z} - w_{,zz} = 0 \tag{29}$$

This differential equation represents the semi-infinite truncated conical rod. So the missing part of the cones from the boundary location z to infinity is modeled by a mechanical system which contains a spring and a damper with frequency-independent coefficients, Eq. (26). The stiffness terms are in inverse proportion to the apex heights of the cones.

It is shown in Ref. [26] that these models, called translational cone models, used for foundation vibration analysis can sufficiently represent body waves in 3D analysis. The same displacement decay low with amplitude in inverse proportion to the distance traveled, starts from the exact Boussinesq solution for the static case, which means that radiation is possible starting at zero frequency.

In a 3D analysis the power of P-waves is maximum directly under the source in the dilatational window and the power of S-waves is concentrated in the shear window. The rest of the halfspace transmits a small part of the radiated power in the far field except for R-waves which propagate horizontally along the surface. From this investigation cone models could be used as a transmitting boundary for body waves in a 3D dynamic analysis of the halfspace medium. The boundary stress vector could be given by Eq. (20) where there are three components in the $\{\sigma\}$, $\{u\}$ and $\{u_{,t}\}$ vectors and

$$[\boldsymbol{D}_K] = \frac{\rho c_p^2}{r} (\boldsymbol{n}\cdot\boldsymbol{r})[\boldsymbol{N}] + \frac{\rho c_s^2}{r} (\boldsymbol{n}\cdot\boldsymbol{r})\{[\boldsymbol{I}] - [\boldsymbol{N}]\} \tag{30}$$

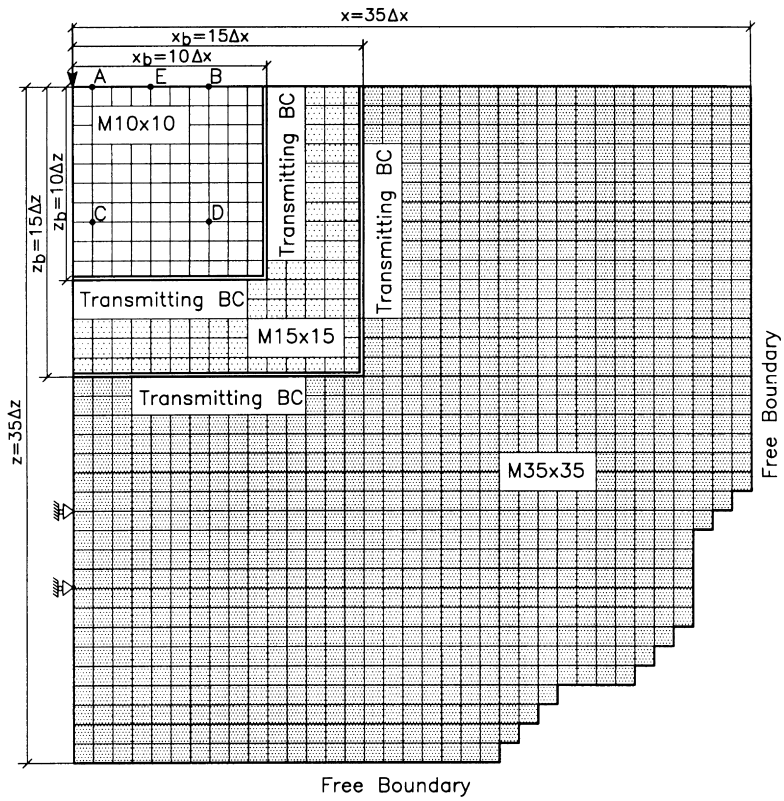

Fig. 3. FE models for the extended mesh and small meshes.

where $\boldsymbol{n}(n_x, n_y, n_z)$, $\boldsymbol{r}(r_x, r_y, r_z)$ and r should be given for each boundary node or each Gauss point and $[\boldsymbol{N}]$ is equal to:

$$[\boldsymbol{N}] = \begin{bmatrix} n_x^2 & n_x n_y & n_x n_z \\ n_y n_x & n_y^2 & n_y n_z \\ n_z n_x & n_z n_y & n_z^2 \end{bmatrix} \tag{31}$$

Consecutive damping matrix $[\boldsymbol{D}_C]$ can be given by Eq. (9) employing Eq. (31) and the vectors $\boldsymbol{n}$ and $\boldsymbol{r}$ for the 3D case.

So the transmitting boundary for body waves can be modeled for each of the degrees-of-freedom (DOFs) at the boundary as a bunch of cones simulated by springs and dashpots attached to the boundary and connected to a rigid base (see Fig. 2). The apexes of the cones are equal to the radius of the hemisphere for a spherical boundary with origin at the source location. For a general boundary shape the apexes of the cones are equal to the distance from the source and the orientation of the vectors $\boldsymbol{n}$ and $\boldsymbol{r}$ should be taken into account. Thus apexes follow from the geometry of the model and source location. The cones will have the same dimensions for all nodal DOFs different from vertical and horizontal cones for surface footings on a halfspace surface [26].

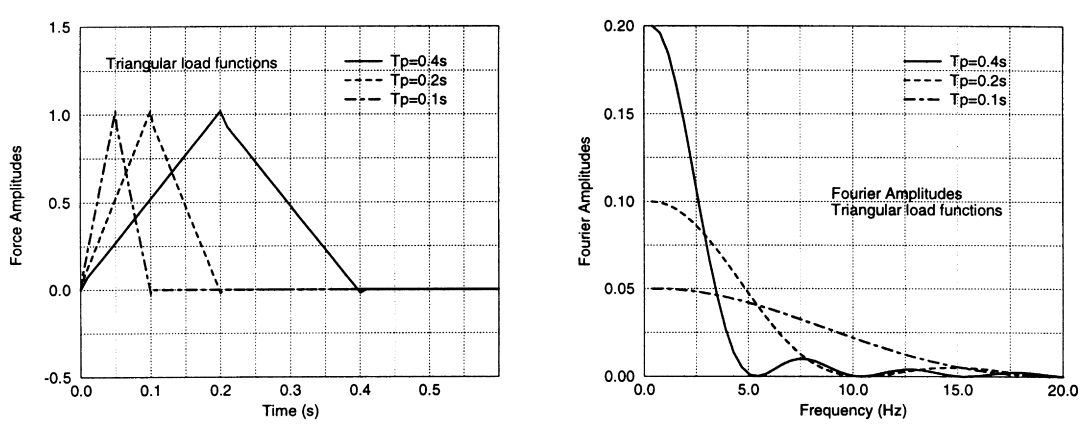

Fig. 4. Delta functions type loads and their Fourier transforms.

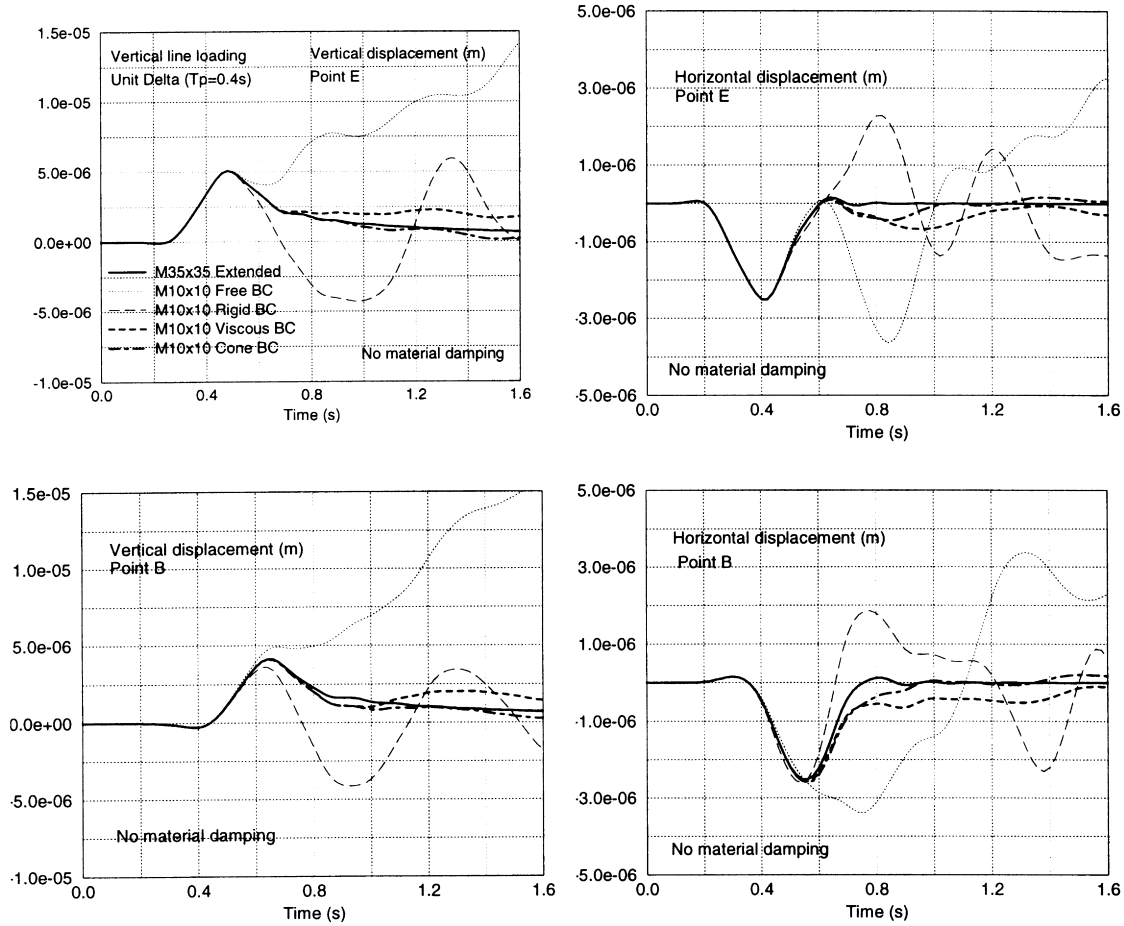

Fig. 5. Comparison of the response at surface points: vertical excitation.

Considering surface waves, as was mentioned before, in a 3D analysis these waves propagate with a cylindrical wavefront. This means that linear cone models could be employed to construct a lateral transmitting boundary. So R-waves decay much more slowly with distance than do body waves. In this case the boundary stress condition is given also by Eq. (29) where $[\boldsymbol{D}_{\mathrm{K}}]$ is equal to:

$$[\boldsymbol{D}_K] = \frac{\rho s c_{\mathrm{R}}^2}{2r}(\boldsymbol{n}\cdot\boldsymbol{r})[\boldsymbol{N}] + \frac{\rho c_{\mathrm{R}}^2}{2r}(\boldsymbol{n}\cdot\boldsymbol{r})\{[\boldsymbol{I}] - [\boldsymbol{N}]\} \tag{32}$$

and $[\boldsymbol{D}_{\mathrm{C}}]$ is given by Eq. (22) with $[\boldsymbol{N}]$ as in Eq. (31).

From the test results it is noted that in modeling the lateral boundary in the 3D case a better approximation should be to use cone models with velocity c_{s} for the out-of-plane motion or SH-waves, instead of the linear cone models with velocity c_{R} used for the components in the in-plane motion [27].

4. Finite element implementation

For 3D analysis, an axisymmetric geometry is preferred here. Axisymmetric analysis can be considered as a variant of 2D analysis, so the derivation of the FE at the boundary is carried out for the 2D quadratic FE. The axisymmetric geometry is taken into account in the integration of the matrices. For simplicity let us consider a rectangular boundary shape. As the quadratic isoparametric FE is chosen to model the field equations because of its proper behavior in dynamic analysis of solids, this element is used to derive the elements at the boundary. The stresses are applied continuously along the boundary surface. Their contribution is assembled at the nodes using 1D FE shape functions. These 1D FEs, which model, for example, the lateral boundary in the plane strain or axisymmetric analysis, are derived as below:

$$\overline{N}_1 = \tfrac{1}{2}(1-\eta) - \tfrac{1}{2}N_3 \tag{33}$$

$$\overline{N}_2 = \tfrac{1}{2}(1+\eta) - \tfrac{1}{2}N_3 \tag{34}$$

$$\overline{N}_3 = 1 - \eta^2 \tag{35}$$

From the FE method the resulting forces acting on the boundary nodes due to the dynamic boundary stresses described in the previous section, are evaluated as:

$$\{F\}^{\mathrm{e}} = -\int_{\Gamma_\infty} [\overline{\boldsymbol{N}}]^{\mathrm{T}}\{\sigma\}\,\mathrm{d}\Gamma \tag{36}$$

where $\{F\}^{\mathrm{e}}$ is the vector of nodal forces at the FE boundary

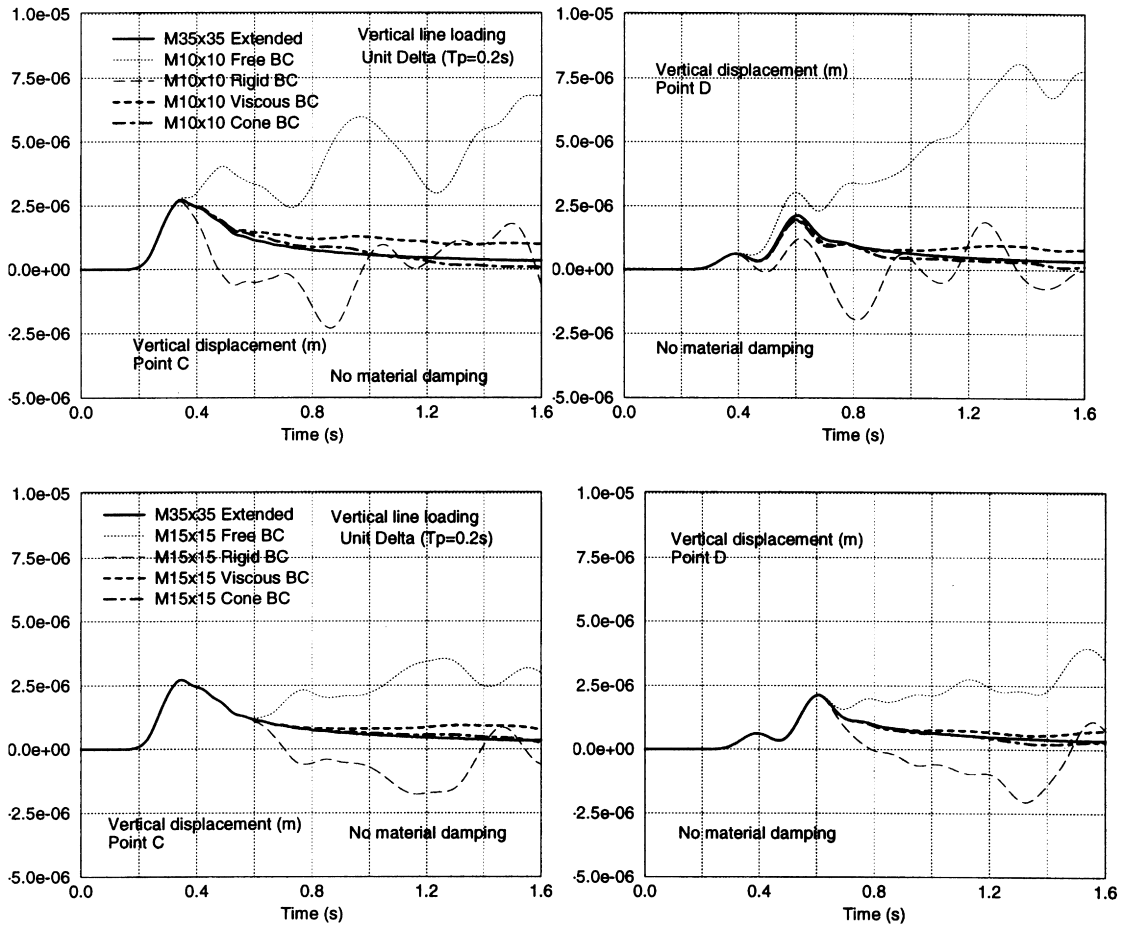

Fig. 6. Comparison of the response for M10 × 10 and M15 × 15: vertical excitation.

and $[\overline{N}]$ is the shape function matrix. Expressing displacements and velocities in terms of nodal values employing the same shape functions and replacing the stress vector from Eq. (20) in general, it follows that:

$$\{F\}^{\mathrm{e}} = -\left(\int_{\Gamma_\infty} [\overline{N}]^{\mathrm{T}}[\boldsymbol{D}_{\mathrm{K}}][\overline{N}]\,\mathrm{d}\Gamma\{u\}^{\mathrm{e}} + \int_{\Gamma_\infty} [\overline{N}]^{\mathrm{T}}[\boldsymbol{D}_{\mathrm{C}}][\overline{N}]\,\mathrm{d}\Gamma\{u_{,t}\}^{\mathrm{e}}\right) \tag{37}$$

The consistent stiffness matrix at the boundary derives as

$$[\boldsymbol{K}]^{\mathrm{e}} = \int_{\Gamma_\infty} [\overline{N}]^{\mathrm{T}}[\boldsymbol{D}_{\mathrm{K}}][\overline{N}]\,\mathrm{d}\Gamma \tag{38}$$

and the consistent damping matrix as

$$[\boldsymbol{C}]^{\mathrm{e}} = \int_{\Gamma_\infty} [\overline{N}]^{\mathrm{T}}[\boldsymbol{D}_{\mathrm{C}}][\overline{N}]\,\mathrm{d}\Gamma \tag{39}$$

The boundary or geometric stiffness and damping matrices for the whole structure–soil-system are obtained by assembling those for the FE boundaries. The consistent matrices locally couple the nodes along the boundary giving a more realistic implementation. These matrices are banded. For comparison lumped stiffness and damping matrices, respectively, are implemented too.

Using another approach, the matrices from the transmitting boundary result as a derivation of the Galerkin formulation. The equations of motion are multiplied by a weight function in the form of a virtual displacement field followed by the integration over the volume and reformulation using the divergence theorem [28].

The equations of motion for the visco-elastic system are

$$[\boldsymbol{M}]_\Omega\{u_{,tt}\}_\Omega + [\boldsymbol{C}]_\Omega\{u_{,t}\}_\Omega + [\boldsymbol{K}]_\Omega\{u\}_\Omega + [\boldsymbol{C}]_{\Gamma_\infty}\{u_{,t}\}_{\Gamma_\infty} + [K]_{\Gamma_\infty}\{u\}_{\Gamma_\infty} = \{\boldsymbol{P}\}_\Omega \tag{40}$$

where $[\boldsymbol{M}]$ is the mass matrix, $[\boldsymbol{C}]$ damping matrices denoted with subscript Ω for internal or material damping and Γ_∞ for radiation damping. $[\boldsymbol{K}]$ stiffness matrices denoting, as before, near- and far-field stiffness. Internal damping is implemented as Rayleigh damping in the form:

$$[\boldsymbol{C}]_\Omega = \alpha_0[\boldsymbol{M}]_\Omega + \beta_0[\boldsymbol{K}]_\Omega \tag{41}$$

where α_0 and β_0 are the so-called Rayleigh damping coefficients. To avoid solution of the eigenvalue problem, these coefficients are related to the modal damping ratio ξ_{s} of the ith mode by relation (42) where ω is the excitation

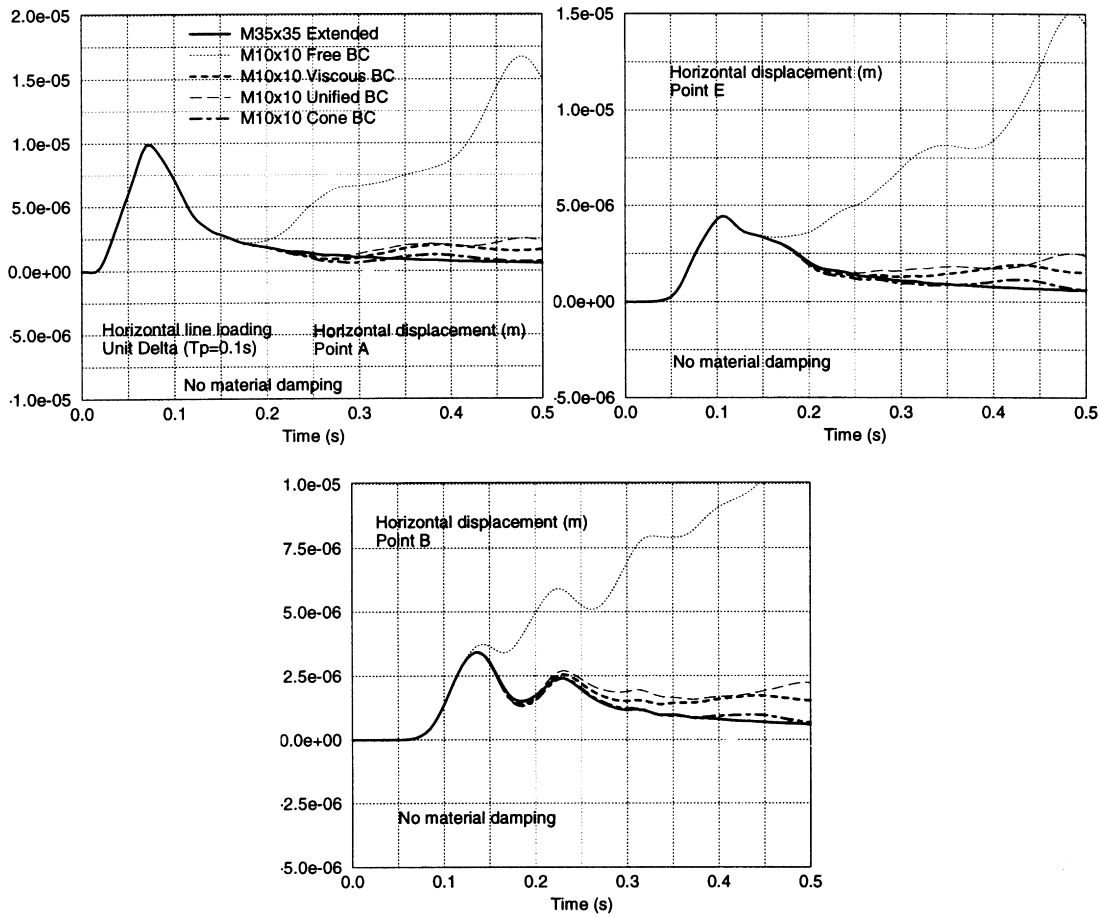

Fig. 7. Comparison of the response at surface points: horizontal excitation.

frequency

$$\xi_s = \frac{\alpha_0 + \beta_0 \omega_i^2}{2\omega_i} \tag{42}$$

The implicit algorithm is preferred to avoid stability problems because in many structural or soil dynamics applications, and especially SSI analysis, important for soft soil conditions, low mode response is of great interest. Wilson θ-method is implemented in the FE programs as an available unconditionally stable one-step algorithm.

5. Computational results

5.1. Plane strain analysis

To see the effectiveness of the transmitting boundary for plane strain analysis numerical experiments are performed in the time domain within the FE models by comparing the results with those of the extended mesh which is taken large enough to prevent reflections from the boundary to the location of interest. As the error due to FE discretization is also present in the extended mesh, the differences are caused by the inherent error from the transmitting boundary. Fig. 3 shows three FE models. Half of the problems are modeled using the symmetry conditions of the loads in dynamic SSI. The models are identified according to the number of FEs in the x- and z-directions.

As the absorbing FEs for body waves are based on the linear cone models they have a stiffness and a damping matrix. At the lateral boundary at a depth approximately equal to χ_R only the viscous boundary is implemented as the boundary stiffness vanishes. A homogeneous, isotropic linear soil halfspace model is chosen with $c_s = 224$ m/s, $\rho = 2000$ kg/m^3 and Poisson's ratio $\nu = 0.25$ which is indicated as M1. The system is excited by a transient line load of Delta type implemented in the program as a triangular pulse. By changing the period T_p, different frequencies are generated, Fig. 4. In choosing the pulse type the frequency range where SSI is of importance for buildings, machine foundations or other man-made loadings is considered.

To see the behavior in the low-frequency interval, a Delta pulse with $T_p = 0.4$ s and unit amplitude is chosen first and applied vertically on the free surface. The FE size is taken $\Delta x = \Delta z = \chi_s/9 = 10$ m where χ_s is the wavelength of the shear waves. The response is investigated at the nodes indicated in Fig. 3. In Fig. 5 the time histories for extended mesh M35 × 35 and M10 × 10 with transmitting BCs are given.

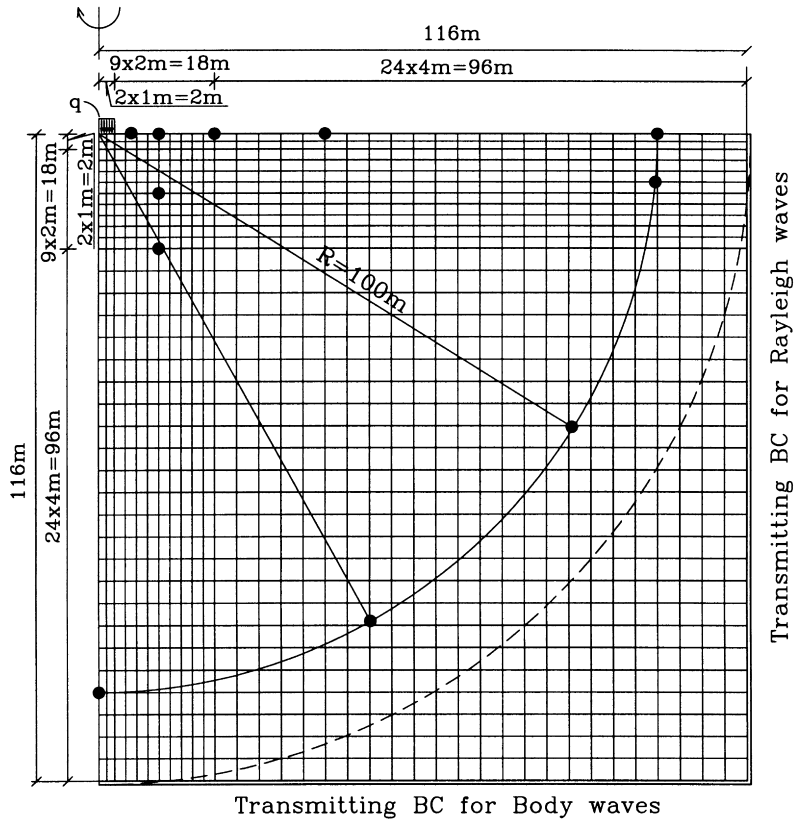

Fig. 8. FE model for axisymmetric analysis.

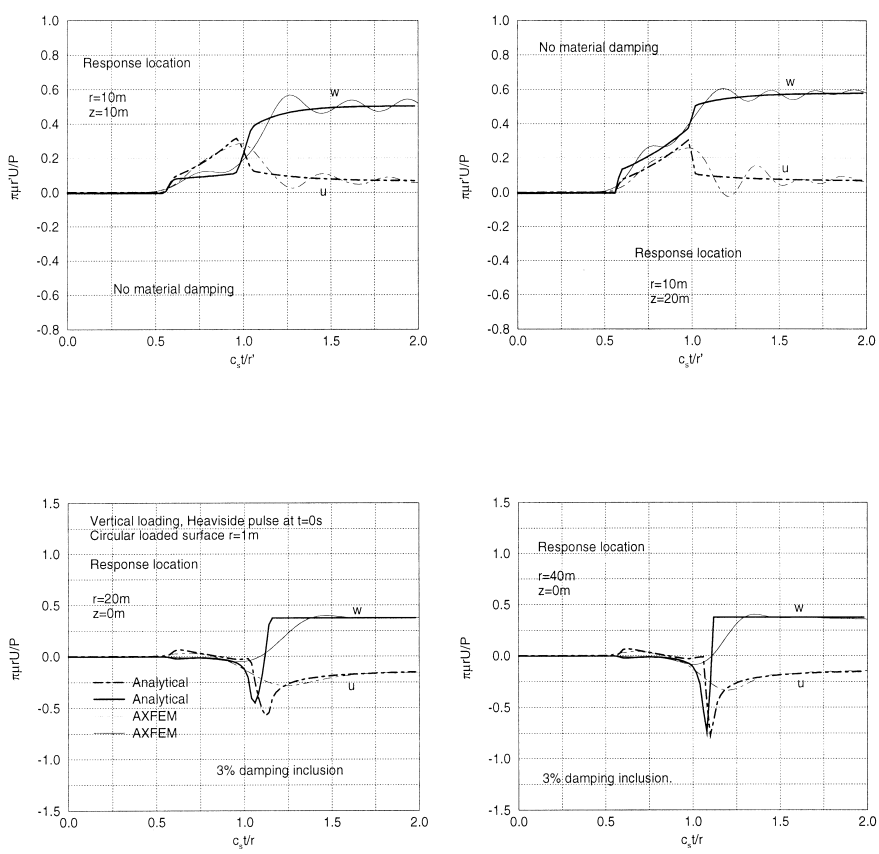

Fig. 9. Comparison of axisymmetric FE computation with analytical solutions: vertical excitations.

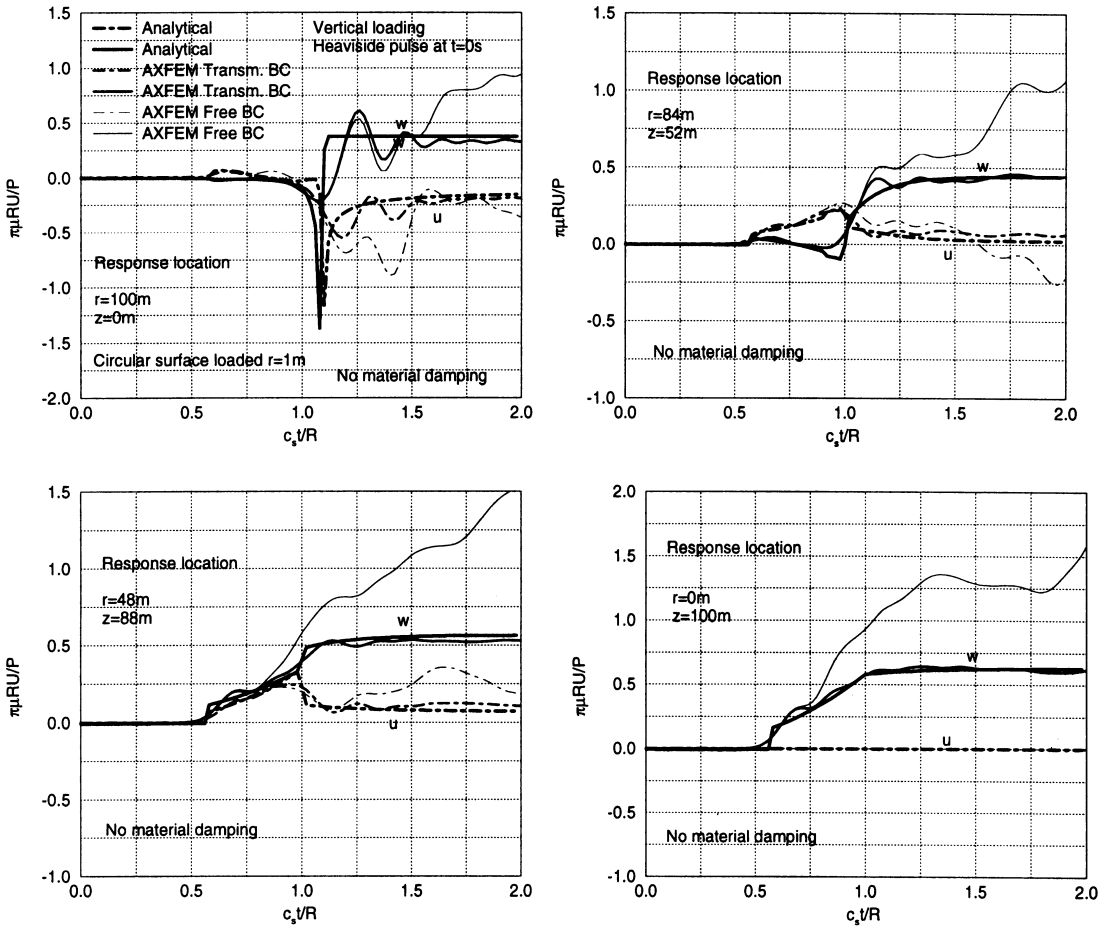

Fig. 10. Comparison of axisymmetric FE computation with BCs and analytical solutions.

From these results, the necessity of absorbing BCs and the great power of the normal impedance boundary are obvious; however, this boundary does not have stiffness and requires load functions of vanishing time history. Also, the FE mesh shows a rigid body movement. By introducing linear cone models for the body waves and viscous boundary considering the angle of incidence for R-waves, the behavior resembles the extended mesh M35 × 35. In the small mesh M10 × 10, the boundary is placed at a distance $x_b = z_b = 1.1\chi_s$. The time duration is long enough to test the boundary capability. Time integration is performed with $\Delta t = 0.02$ s. To clearly see the effect of radiation damping and absorbing of the energy from boundaries, no material damping is included in these soil models. In Fig. 6 the response at nodes C and D for M10 × 10 and M15 × 15 are given for a Delta load function with $T_p = 0.2$ s. So the frequencies generated in the system are doubled but the element size is maintained as in the previous computations corresponding to $\Delta x = \Delta z = \chi_s/4.5$, which gives good behavior. By comparing the results for the vertical displacement components in the same nodes the effect of the boundary location can be investigated. The boundary corresponds to a distance of $x_b = z_b = 2.2\chi_s$ for M10 × 10 and $x_b = z_b = 3.4\chi_s$ for M15 × 15. In both cases the boundary is far enough from the source. Time integration is carried out with $\Delta t = 0.01$ s. In these graphics the new boundary is called a cone boundary.

To check the behavior of the local boundary for other soil material, another model M2 is taken with material properties $c_s = 200$ m/s, $\rho = 1800$ kg/m^3 and $\nu = 0.4$. The pulse is sharper with $T_p = 0.1$ s. As both vertical and horizontal load components are important in dynamic SSI, the line load horizontal excitation is considered now. The FE size is taken $\Delta x = \Delta z = 4$ m $= \chi_s/5$. The boundary corresponds to $x_b = z_b = 2\chi_s$ from the source. The total time of investigation is $t = 0.5$ s to avoid boundary effect from M35 × 35 to the selected nodes. The time integration is carried out with $\Delta t = 0.005$ s. During these analysis lumped and consistent boundary matrices were used and no important difference in accuracy was noticed. The behavior at nodes A, E and B for the horizontal components is given in Fig. 7. There is no doubt that the new boundary employing linear cones is more accurate than the viscous or unified one. The stiffness of the unbounded soil domain, which is missing in the viscous boundary can be approximated by the linear cone boundary.

5.2. Axisymmetric analysis

For symmetric loading the FE nodes are circles having

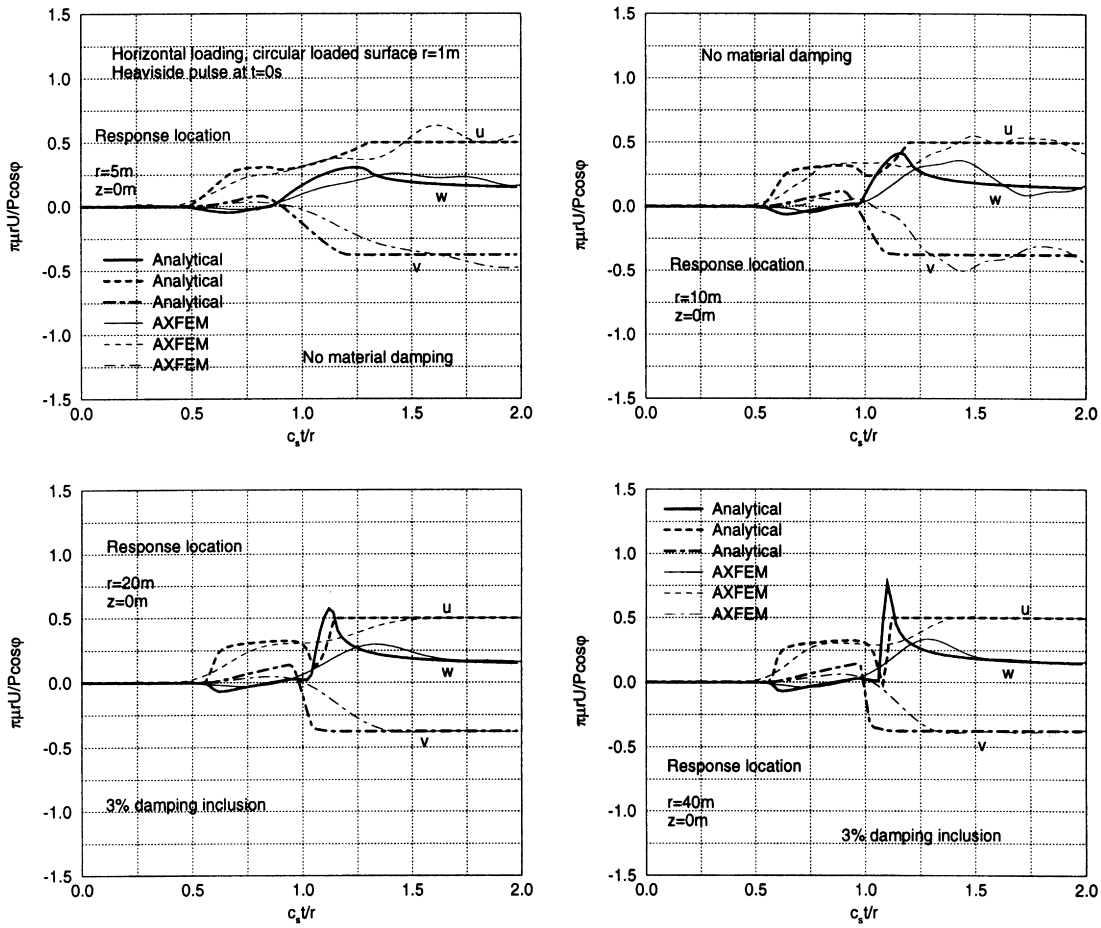

Fig. 11. Comparison of axisymmetric FE computation with analytical solutions: horizontal excitation.

two DOFs. In formulating the matrices for the field and boundary equations the radial coordinate of each Gauss point is included in the integration. Before testing the new boundary for vertical loading the FE program was tested first with analytical solutions or transient 3D Lamb's solution, a classical approach [29]. The FE model is given in Fig. 8 with material properties of M1. A Heaviside impulse load at $t = 0$ s is considered acting on a circular surface of $r = 1$ m in such a way that the total load $P = 1$ kN. A predicted FE discretization and boundary location is difficult in this case. From the experience the FE model is built with $\Delta x = \Delta z \leq 4$ m and the boundary is placed at $x_b = z_b = 116$ m from the axis of symmetry to include nodes placed at a circle of radius $R = 100$ m from the origin where the closed form solution is known and the boundary effect can be investigated.

This type of load with constant time history is a test for the new arranged boundary which models approximately also the stiffness of the unbounded soil domain. The locations denoted with bold dots are investigated. Some of them are shown here specifying the coordinates of the locations. The displacement response for vertical w and horizontal u components are given in nondimensionalized form in Fig. 9. In those graphics in the ordinate, r or r' is the shorter distance of the location from the origin of the model. U is used to symbolize both components of displacements. The variation is given versus nondimensionalized time till the value 2. So the response is not effected from the BCs. The closed-form solutions are derived by use of the integral transform technique introducing the displacement potentials which are expanded into the Fourier series along the azimuthal direction. The Laplace transform is taken with respect to time. The Hankel transform for the radial direction decouples the wave field into the in-plane and the out-of-plane motions. Application of the BCs determines the solution in the transformed domain. The inverse process proceeds first with respect to the Laplace parameter performed by contour integration on the complex plane. Then the inverse Hankel transform in closed form leads the time space domain solutions in the form of the semi-infinite integrals.

When no material damping is included, some oscillations can be observed originating from mesh discretization which filters high frequencies in the system. This can be improved by introducing some low damping.

To test the new boundary based on cones for body waves and linear cones for surface waves, the response at the nodes near the boundary in a circle with a radius $R = 100$ m from

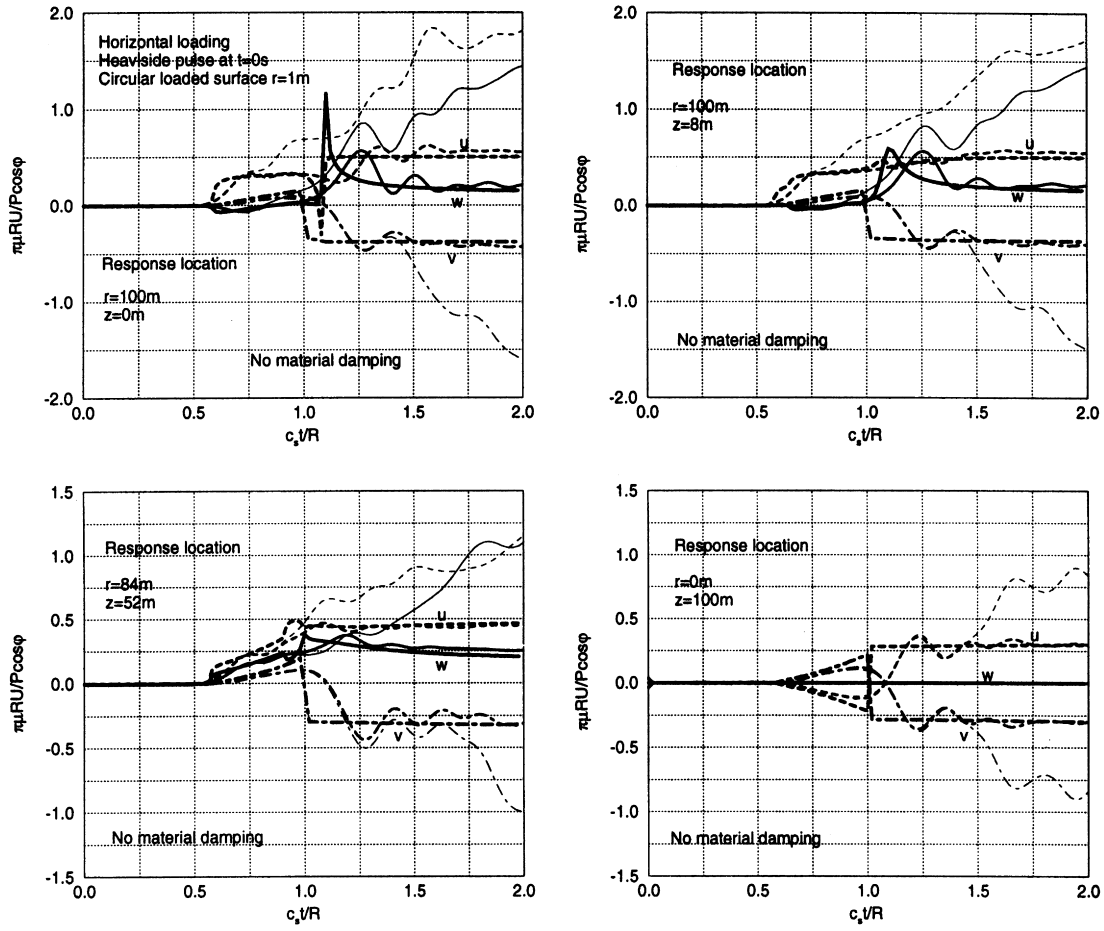

Fig. 12. Comparison of axisymmetric FE computation with BCs and analytical solutions.

the origin is investigated. The results seem to be quite accepted considering FE approximation from the mesh discretization. No material damping was introduced in these computations (Fig. 10).

For horizontal loading or nonsymmetric loading it is assumed that the applied load and the nodal circle displacements are expanded into Fourier series. Utilizing the orthogonal properties of the harmonic functions the 3D analysis is reduced to a series of uncoupled 2D analyses. The number of 2D problems is determined by the number of harmonics required to represent the nonsymmetric loading. The nonsymmetrically loaded halfspace differs from the symmetric case by the addition of a horizontally polarized shear wave. In this case each nodal circle has three DOFs. The horizontally polarized SH-wave is considered to decay in the same way as body waves for halfspace models and could be considered to decay as surface waves for models with layers over a rigid base.

If the distributed load were complicated, several loading harmonic terms would be necessary. For the constant load distribution considered, one harmonic is sufficient. First, the response at near-field nodes for the three DOFs u, w, and v is compared with the closed-form solutions in Fig. 11. The ordinate shows that one harmonic is taken for the load and this ratio is valid for horizontal u and vertical w components. For the tangential component which for convenience is denoted as v, $P \sin \phi$ should be understood for the load.

The boundary effect in this case is investigated from the response at the near-boundary nodes as for the vertical loading. These results are given in Fig. 12 where, the capacity of the new transmitting boundary for an absorbing body and surface waves in a 3D dynamic analysis can be seen.

6. Discussions and conclusions

Radiation and BCs can be interpreted as constitutive equations for the interaction forces between the near and far fields. It is reported by Meek and Wolf [30] that for frequencies near zero R-waves transmit most of the power. As frequency increases the importance of R-waves compared to P- and S-waves decreases rapidly. In the high-frequency intervals almost all the energy is transmitted by P-waves. The explanation is that χ_R being inversely proportional to the excitation frequency ω determines the height of the cylindrical surface at infinity through which the energy from R-waves passes. When ω increases, the depth χ_R of the cylindrical layer decreases. This means that in the

high-frequency range, linear cone models for 2D and cone models for 3D can be used as transmitting boundary for all the waves when they are placed at a certain distance from the source. In this case, the models could be formulated in cylindrical and spherical coordinates, respectively, and the transmitting boundary can also be interpreted as applying to each of the DOFs at the boundary, first-order operators of BCs formulated from Ref. [15] for acoustic media.

BCs formulated here can be considered as DA ones and a generalization of the viscous boundary. In Ref. [21] it was found that the MD boundary or viscous boundary satisfies the Kreiss criterion which determines how well the initial boundary-value problem has been posed for first-order linear hyperbolic systems. However, this criterion is not satisfied for zero frequency, or static case. The reason for this exception is the fact that any differential operator is zero at frequency zero if this operator does not have a constant term. This can be seen as a kind of incompatibility between the initial data and the BCs. A modification was proposed in Ref. [21] by writing the operators in the form:

$$\cos\alpha\frac{\partial}{\partial t} + c_{\mathrm{p}}\frac{\partial}{\partial x} + \kappa \tag{43}$$

where $\kappa > 0$. If the stiffness term from the linear cones or cones is used instead of κ then the Kreiss criterion is satisfied also for the static case, which makes the operator DA. The purpose is to increase the accuracy of the wave propagation as $t \to \infty$. So the formulated boundaries for 2D or 3D analysis represent an attempt to construct a local stiffness matrix for the unbounded soil domain.

No improvements were noticed when employing a unified viscous boundary compared to the standard viscous boundary, but if the unified dashpot is combined with the stiffness of the cones a tendency for improvement was reported.

Acknowledgements

This work was supported by the Department of Structural Engineering & Materials at the Technical University of Denmark. The author thanks Profs H. Takemiya and S. Krenk for their contributions through discussions and Dr L.O. Nielsen for his help in developing the FE code.

References

[1] Sandler IS. A method of successive approximations for structure interaction problems. Computation methods for infinite domain media–structure interaction, AMD, 46. ASME, 1981 (p. 67–82).
[2] Wolf JP. Soil–structure-interaction analysis in time domain. Englewood Cliffs, NJ: Prentice-Hall, 1988.
[3] Givoli D. Non-reflecting boundary conditions. Journal of Computational Physics 1991;94:1–29.
[4] Lysmer J, Kuhlemeyer RL. Finite dynamic model for infinite media. Journal of the Engineering Mechanics Division, ASCE 1969:859–77.
[5] White W, Valliappan S, Lee IK. Unified boundary for finite dynamic models. Journal of the Engineering Mechanics Division, ASCE 1977;103(EM5):949–64.
[6] Akiyoshi T. Compatible viscous boundary for discrete models. Journal of the Engineering Mechanics Division, ASCE 1978; 104(EM5):1253–66.
[7] Smith WD. A nonreflecting plane boundary for wave propagation problems. Journal of Computational Physics 1974;15:492–503.
[8] Kunar RR, Marti J. A non-reflecting boundary for explicit calculations. Computational Methods for Infinite Domain Media–Structure Interaction, AMD, 46. ASME, 1981 (p. 183–204).
[9] Lindman EL. Free-space boundary conditions for the time dependent wave equation. Journal of Computational Physics 1975;18:66–78.
[10] Engquist B, Majda A. Absorbing boundary conditions for the numerical simulation of waves. Mathematics of Computation 1977;31: 629–51.
[11] Clayton RW, Engquist B. Absorbing boundary conditions for acoustic and elastic wave equation. Bulletin of the Seismological Society of America 1977;67(6):1529–41.
[12] Clayton RW, Engquist B. Absorbing boundary conditions for wave-equation migration. Geophysics 1980;45:895–904.
[13] Reynolds AC. Boundary conditions for the numerical solution of wave propagation problems. Geophysics 1978;43:1099–110.
[14] Cohen M, Jennings PC. Silent boundary methods for transient analysis. In: Belytschko T, Hughes TJR, editors. Computational methods for transient analysis. Amsterdam: Elsevier, 1983. p. 301–60.
[15] Bayliss A, Turkel E. Radiation boundary conditions for wave-like equations. Communications of Pure Applied Mathematics 1980; 33:707–25.
[16] Liao ZP, Wong HL. A transmitting boundary for the numerical simulation of elastic wave propagation. Soil Dynamics and Earthquake Engineering 1984;3:174–83.
[17] Cheng N, Cheng CH. Relationship between Liao and Clayton–Engquist absorbing boundary conditions: acoustic case. Bulletin of the Seismological Society of America 1995;85:954–6.
[18] Underwood P, Geers TL. Doubly asymptotic, boundary element analysis of dynamic soil–structure interaction. International Journal of Solids and Structure 1981;17:687–97.
[19] Keys RG. Absorbing boundary conditions for acoustic media. Geophysics 1985;50:892–902.
[20] Higdon RL. Numerical absorbing boundary conditions for the wave equation. Mathematics of Computations 1987;49:65–90.
[21] Higdon RL. Radiation boundary conditions for elastic wave propagation. SIAM Journal of Numerical Analysis 1990;27:831–70.
[22] Higdon RL. Absorbing boundary conditions for elastic waves. Geophysics 1991;56:231–41.
[23] Wolf JP, Song C. Finite-element modeling of unbounded media. New York: Wiley, 1996.
[24] Kausel E. Local transmitting boundaries. Journal of Engineering Mechanics, ASCE 1988;114:1011–27.
[25] Graff KF. In: Woods LC, Wittrick WH, Cullen AL, editors. Wave motions in elastic solids. Oxford engineering science series 1975.
[26] Wolf JP. Foundation vibration analysis using simple physical models. Englewood Cliffs, NJ: Prentice-Hall, 1994.
[27] Kellezi L. Dynamic soil–structure-interaction. Transmitting boundary for transient analysis. PhD thesis, Department of Structural Engineering and Materials, Technical University of Denmark, S.R. No. 50, 1998.
[28] Krenk S, Kellezi L, Nielsen SRK, Kirkegaard PH. Finite elements and transmitting boundary conditions for moving loads. Eurodyn'99, Prague, Czechoslovakia.
[29] Takemiya H, Steinfeld B. Transient 3D Lamb's solution by classical approach and direct boundary element method. Structural dynamics. Eurodyn'93, Balkema, Rotterdam.
[30] Meek JW, Wolf JP. Why cone models can represent the elastic half-space. Earthquake Engineering and Structural Dynamics 1993; 22:759–72.